16-Day Roadmap: Agentic AI + Generative AI

Build strong practical expertise in Generative AI and Agentic AI with hands-on, production-ready systems using LLMs, RAG, and autonomous agents.

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COURSE OVERVIEW

Learning Journey

01

Foundation

AI fundamentals, NLP evolution, and transformers

02

Generative AI

Prompt engineering, LLM APIs, and RAG systems

03

Agentic AI

Agent architecture, multi-agent systems, and frameworks

04

Production

Deployment, evaluation, and capstone project

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DAYS 1-4

AI Foundations & LLM Basics

Day 1: Core Concepts

AI vs ML vs DL, supervised & unsupervised learning with hands-on ML examples

Day 2: NLP Evolution

Transformers, embeddings, and similarity search techniques

Day 3: Prompt Engineering

Prompt templates, hallucination handling, and optimization strategies

Day 4: LLM APIs

OpenAI, Gemini, Azure integration, token usage, and cost control

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DAYS 5-7

RAG & Structured Outputs

Day 5: RAG Fundamentals

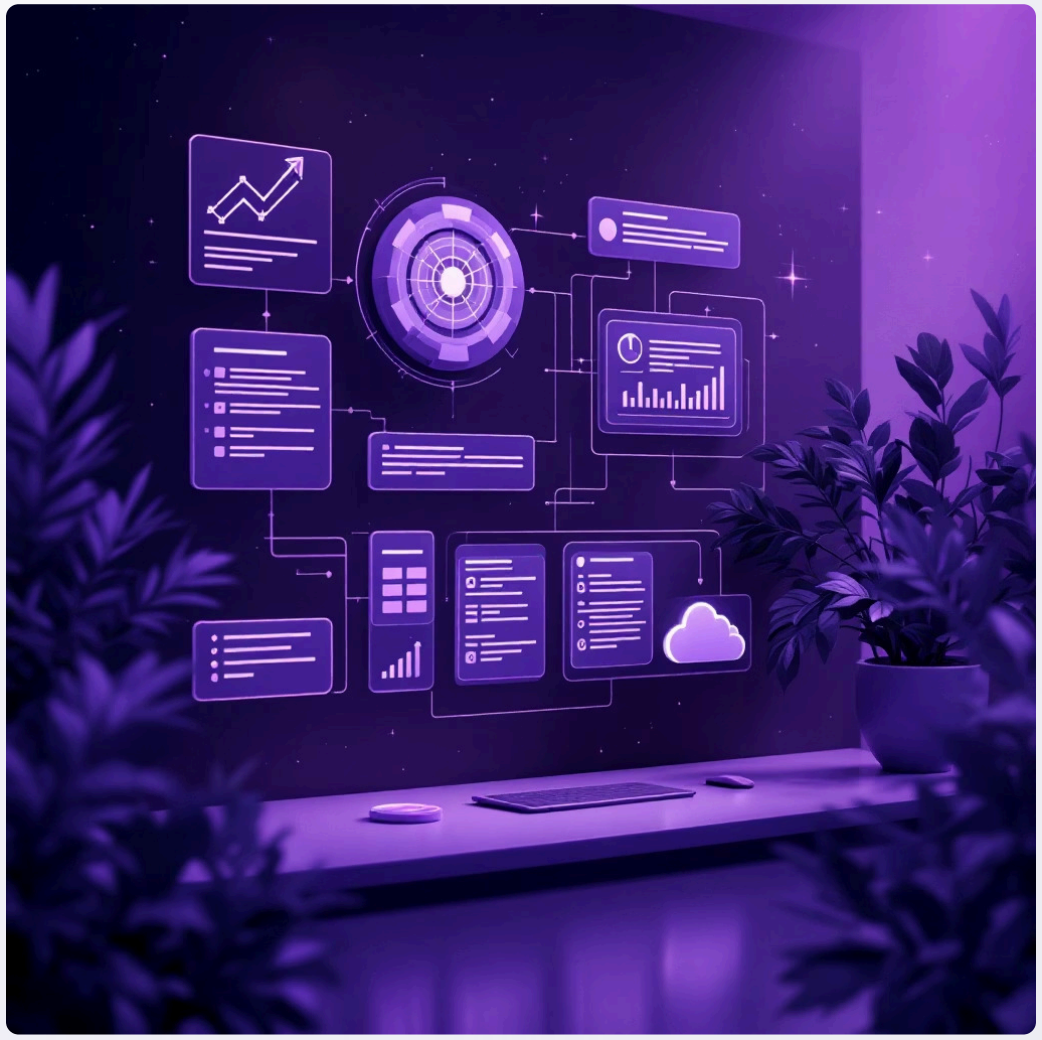
- Embeddings and vector databases
- Retrieval mechanisms
- Basic RAG architecture

Day 6: Advanced RAG

- Reranking strategies
- Metadata filtering
- Evaluation metrics

Day 7: Structured Outputs

- JSON generation
- Text-to-SQL conversion
- Document processing



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DAYS 8-10

Agentic AI Architecture

Day 8: Agent Concepts

Agent architecture, tool usage, and autonomous decision-making

Day 9: Multi-Step Reasoning

Tool calling, error handling, and complex workflows

Day 10: Multi-Agent Systems

Role-based collaboration and agent coordination

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DAYS 11-13

Frameworks & Production Systems

1	2	3
Agent Frameworks	**Agentic RAG**	**Safety & Evaluation**
LangChain, LangGraph, and AutoGen with memory management	Self-querying agents and advanced retrieval systems	Observability, guardrails, and comprehensive evaluation metrics

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DAYS 14-16

Deployment & Capstone

Day 14: Production Deployment

Streamlit and FastAPI integration with latency optimization

Day 15: Capstone Project

Build an end-to-end agentic system from scratch

Day 16: Final Review

System design patterns and interview preparation

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INDUSTRY APPLICATIONS

Finance & Healthcare Use Cases

Banking & Finance

AI financial analyst agents, credit risk analysis, and fraud investigation assistants

Healthcare

Clinical documentation agents, medical research summarization, and patient query assistants

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INDUSTRY APPLICATIONS

Retail, Legal & HR Solutions

Retail & E-commerce

Personalized recommendation agents and demand forecasting assistants

Legal

Contract analysis agents and compliance monitoring systems

HR & Talent

Resume screening agents and interview scheduling bots

Manufacturing

Predictive maintenance agents and quality inspection copilots

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INDUSTRY APPLICATIONS

Operations & Analytics

Data & Analytics

AI data analyst agents with KPI generation and reporting automation

Customer Support

Autonomous support agents with RAG over company knowledge bases

Supply Chain

Route optimization agents and demand planning assistants

IT & DevOps

Incident analysis agents and log summarization for root cause analysis

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